

传感器集成系统习题详细解答与总结

Intro to Sensor-Integrated Systems — Problem Solutions & Knowledge Summaries

Tutorials 1–6, 9, 10, 11

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Tutorial 1: The General Measurement System & Static Characteristics

背景知识 (Background)

测量系统的精度定义为测量值与真实值的接近程度。系统误差 $E = O - I$ (输出值减去真实输入值)。理想直线方程: $O_{\text{ideal}} = KI + a$, 其中灵敏度 (斜率) $K = \frac{O_{\text{max}} - O_{\text{min}}}{I_{\text{max}} - I_{\text{min}}}$, 零偏 (截距) $a = O_{\text{min}} - KI_{\text{min}}$ 。

Q1-1: 理想直线与灵敏度 (Ideal Straight Line & Sensitivity)

(a) Force sensor

A force sensor has input range $0 \rightarrow 100 \text{ N}$ and output range $1 \text{ V} \rightarrow 5 \text{ V}$. Find the sensitivity K and zero offset a .

解题过程:

$$K = \frac{O_{\text{max}} - O_{\text{min}}}{I_{\text{max}} - I_{\text{min}}} = \frac{5 - 1}{100 - 0} = \frac{4}{100} = 0.04 \text{ V/N}$$
$$a = O_{\text{min}} - K \cdot I_{\text{min}} = 1 - 0.04 \times 0 = 1 \text{ V}$$

因此理想直线方程为 $O = 0.04I + 1$ (单位: V, N)。

(b) Current transmitter

A current transmitter has input range $0 \rightarrow 100 \text{ kPa}$ and output range $4 \text{ mA} \rightarrow 20 \text{ mA}$. Find K and a .

$$K = \frac{20 - 4}{100 - 0} = 0.16 \text{ mA/kPa}, \quad a = 4 - 0.16 \times 0 = 4 \text{ mA}$$

Q1-2: 非线性误差 (Non-linearity Error)

Pressure sensor non-linearity

A pressure sensor has output span 5 V . At 4 bar , the actual output is 0.20 V above the ideal line. Calculate (1) the non-linearity error at 4 bar ; (2) the non-linearity as %f.s.d.

解:

1. 非线性误差: $N(4 \text{ bar}) = 0.20 \text{ V}$

2. 占满偏的百分比:

$$\% \text{f.s.d.} = \frac{\hat{N}}{O_{\text{max}} - O_{\text{min}}} \times 100\% = \frac{0.20}{5} \times 100\% = 4\%$$

Q1-3: 干扰输入与修正输入 (Interfering & Modifying Inputs)

Complete static model

A sensor obeys $O = (K + K_M I_M)I + a + K_I I_I$. Given: $K = 0.04 \text{ V/N}$, $K_M = 0.001 \text{ V/(N} \cdot ^\circ\text{C)}$, $K_I = 0.02 \text{ V/}^\circ\text{C}$, $I_M = +10^\circ\text{C}$, $I_I = +5^\circ\text{C}$. Find the actual output at $I = 50 \text{ N}$.

$$O = (0.04 + 0.001 \times 10) \times 50 + 1 + 0.02 \times 5 = 0.05 \times 50 + 1 + 0.1 = 2.5 + 1 + 0.1 = 3.6 \text{ V}$$

Q1-4: ADC 分辨率与量化误差 (ADC Resolution)

n-bit ADC

An 8-bit ADC has input range $0 \rightarrow 5 \text{ V}$. Calculate the resolution ΔV and the resolution as %f.s.d.

$$\Delta V = \frac{V_{\max} - V_{\min}}{2^n} = \frac{5}{256} \approx 19.53 \text{ mV}$$
$$\% \text{f.s.d.} = \frac{1}{2^n} \times 100\% = \frac{1}{256} \times 100\% \approx 0.39\%$$

Tutorial 1 知识点总结

- 理想直线: $O = KI + a$, 斜率 K 即灵敏度, 截距 a 为零偏。
- 非线性误差: $N(I) = O(I) - (KI + a)$; 以满偏百分比 $\hat{N}/(O_{\max} - O_{\min}) \times 100\%$ 表示。
- 干扰输入 I_I : 改变零偏 (纵向平移), 不改变斜率。
- 修正输入 I_M : 改变灵敏度 (斜率), 不改变截距。
- ADC 分辨率: $\Delta V = V_{\text{span}}/2^n$; 迟滞误差 $\% \text{f.s.d.} = H/(O_{\max} - O_{\min}) \times 100\%$ 。

Tutorial 2: Systematic Error, Error Propagation & Feedback

背景知识

串联系统总灵敏度: $K_{\text{sys}} = K_1 K_2 \cdots K_n$ 。误差带 $\pm h$ 对应方差 $\sigma^2 = h^2/3$ (均匀分布)。方差传播递推: $\sigma_{\text{out},i}^2 = K_{i+1}^2 \sigma_{\text{out},i-1}^2 + h_{i+1}^2/3$ 。

Q2-1: 串联系统误差传播 (Series System Error Propagation)

Three-element series system

A measurement system has three elements with $K_1 = 2$, $K_2 = 5$, $K_3 = 0.1$ and accuracy bands $h_1 = \pm 0.5\%$ f.s.d., $h_2 = \pm 1\%$ f.s.d., $h_3 = \pm 0.5\%$ f.s.d.. Find the total output standard deviation.

方差传播逐级计算:

第 1 级输出方差 (初始输入无误差, $\sigma_{\text{in},1}^2 = 0$):

$$\sigma_1^2 = \frac{h_1^2}{3}$$

第 2 级输出方差:

$$\sigma_2^2 = K_2^2 \sigma_1^2 + \frac{h_2^2}{3}$$

第 3 级输出方差 (最终输出):

$$\sigma_3^2 = K_3^2 \sigma_2^2 + \frac{h_3^2}{3}$$

总输出标准差: $\sigma_E = \sqrt{\sigma_3^2}$ 。

Q2-2: 负反馈高增益近似 (High-Gain Negative Feedback)

Closed-loop system

A closed-loop system has forward gain $K_{\text{fwd}} = 1000$ and feedback element $K_{\text{fb}} = 0.1$. Find the closed-loop sensitivity K_{cl} .

精确公式:

$$K_{\text{cl}} = \frac{K_{\text{fwd}}}{1 + K_{\text{fwd}} K_{\text{fb}}} = \frac{1000}{1 + 1000 \times 0.1} = \frac{1000}{101} \approx 9.90$$

高增益近似 ($K_{\text{fwd}} K_{\text{fb}} \gg 1$):

$$K_{\text{cl}} \approx \frac{1}{K_{\text{fb}}} = \frac{1}{0.1} = 10$$

结论: 闭环灵敏度仅取决于反馈元件, 与前向通道参数漂移无关。

Q2-3: 非理想系统综合误差分析 (Comprehensive Statistical Error Analysis)

Nonlinear flow meter

A flow meter uses $Q = C\sqrt{\Delta P/\rho}$. Nominal: $\Delta P = 1000$ Pa, $\bar{\rho} = 1.2$ kg/m³, $C = 0.5$. Random variations: $\sigma_{\Delta P} = 10$ Pa, $\sigma_{\rho} = 0.05$ kg/m³. Calculate mean output $\bar{Q}$ and output standard deviation σ_Q .

Track 1 —均值计算:

$$\bar{Q} = C\sqrt{\frac{\Delta P}{\bar{\rho}}} = 0.5\sqrt{\frac{1000}{1.2}} = 0.5 \times 28.87 \approx 14.43 \text{ m}^3/\text{s}$$

Track 2 —方差传播 (偏导数法):

偏导数 (在均值点处求值):

$$\frac{\partial Q}{\partial \Delta P} = \frac{C}{2\sqrt{\Delta P \cdot \rho}} = \frac{0.5}{2\sqrt{1000 \times 1.2}} = \frac{0.5}{69.28} \approx 7.22 \times 10^{-3}$$

$$\frac{\partial Q}{\partial \rho} = -\frac{C\sqrt{\Delta P}}{2\rho^{3/2}} = -\frac{0.5 \times \sqrt{1000}}{2 \times 1.2^{3/2}} \approx -6.01$$

方差合成:

$$\sigma_Q^2 = \left(\frac{\partial Q}{\partial \Delta P}\right)^2 \sigma_{\Delta P}^2 + \left(\frac{\partial Q}{\partial \rho}\right)^2 \sigma_{\rho}^2 = (7.22 \times 10^{-3})^2 \times 10^2 + (6.01)^2 \times (0.05)^2 \approx 0.00521 + 0.0902 \approx 0.0954$$

$$\sigma_Q \approx 0.309 \text{ m}^3/\text{s}$$

工程诊断: σ_{ρ} 项贡献占主导 $\Rightarrow$ 应增加温度/压力传感器对密度进行动态补偿 (“温压补偿”)。

Tutorial 2 知识点总结

- 误差带转方差: $\sigma^2 = h^2/3$ (均匀分布假设)。
- 串联方差传播: 每级误差被后续灵敏度系数平方放大并累积。
- 第一级规律: 第一级灵敏度 K_1 永远不会放大本级误差!
- 高增益负反馈: $K_{cl} \approx 1/K_{fb}$, 系统性能仅取决于反馈环路的精度。
- 双轨道算法: 均值误差 ($\bar{E}$) 用均值代入计算; 随机误差 (σ_E) 用偏导数传播法计算。
- 优化方向: $\bar{E}$ 大 $\Rightarrow$ 改软件算法; σ_E 大 $\Rightarrow$ 增加辅助传感器做动态补偿。

Tutorial 3: Dynamic Response — Step & Frequency

背景知识

一阶系统阶跃响应:

$$\Delta O(t) = K \cdot \Delta I \cdot (1 - e^{-t/\tau})$$

当 $t = \tau$ 时, 输出达稳态变化量的 63.2%; 当 $t = 5\tau$ 时基本达到稳态。

二阶系统 (欠阻尼 $\xi < 1$) 阶跃响应:

$$\Delta O(t) = K \cdot \Delta I \cdot \left[1 - e^{-\xi\omega_n t} \left(\cos \omega_d t + \frac{\xi}{\sqrt{1-\xi^2}} \sin \omega_d t \right) \right], \quad \omega_d = \omega_n \sqrt{1-\xi^2}$$

一阶系统频率响应:

$$|G(j\omega)| = \frac{K}{\sqrt{1+(\omega\tau)^2}}, \quad \phi = -\arctan(\omega\tau)$$

二阶系统频率响应:

$$|G(j\omega)| = \frac{K}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[\frac{2\xi\omega}{\omega_n}\right]^2}}$$

Q3-1: 一阶系统时间常数 (First-Order Time Constant)

Thermocouple step test

A thermocouple is placed in a fluid at 100°C (initially at 20°C). After 10s, its reading is 76.3°C . Find the time constant τ .

稳态变化量 $\Delta I = 100 - 20 = 80^\circ\text{C}$; $t = 10\text{ s}$ 时输出变化量 $\Delta O = 76.3 - 20 = 56.3^\circ\text{C}$ 。

$$\frac{\Delta O}{\Delta I} = 1 - e^{-t/\tau} \implies \frac{56.3}{80} = 1 - e^{-10/\tau}$$

$$e^{-10/\tau} = 1 - 0.7038 = 0.2963 \implies \frac{10}{\tau} = -\ln(0.2963) = 1.216 \implies \tau = \frac{10}{1.216} \approx 8.22\text{ s}$$

Q3-2: 二阶系统自然频率与阻尼比 (Natural Frequency & Damping Ratio)

Force transducer design

A force transducer has mass $m = 0.1\text{ kg}$, spring stiffness $k = 10^4\text{ N/m}$, damper $\lambda = 20\text{ N} \cdot \text{s/m}$. Find ω_n , ξ , and settling time T_s .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{10^4}{0.1}} = \sqrt{10^5} \approx 316.2\text{ rad/s}$$
$$\xi = \frac{\lambda}{2\sqrt{km}} = \frac{20}{2\sqrt{10^4 \times 0.1}} = \frac{20}{2 \times 31.62} = \frac{20}{63.25} \approx 0.316$$

调节时间 (欠阻尼):

$$T_s \approx \frac{5}{\xi\omega_n} = \frac{5}{0.316 \times 316.2} \approx \frac{5}{99.9} \approx 0.050\text{ s}$$

Q3-3: 开环动态补偿 (Open-Loop Dynamic Compensation)

Lead-lag compensation

A sensor has transfer function $G_s(s) = \frac{1}{1 + \tau_s s}$ with $\tau_s = 1$ s. Design a compensation circuit $G_c(s)$ to reduce the effective time constant to $\tau_f = 0.01$ s.

零极点对消策略：令补偿电路在 $s = -1/\tau_s$ 处产生一个零点，在 $s = -1/\tau_f$ 处引入新极点：

$$G_c(s) = \frac{1 + \tau_s s}{1 + \tau_f s} = \frac{1 + s}{1 + 0.01s}$$

总传递函数：

$$G_{\text{total}}(s) = G_s(s) \cdot G_c(s) = \frac{1}{1 + s} \cdot \frac{1 + s}{1 + 0.01s} = \frac{1}{1 + 0.01s}$$

等效时间常数从 1 s 降至 0.01 s，响应速度提升 100 倍。

Tutorial 3 知识点总结

- **时间常数 τ** ：一阶系统响应速度的核心指标。 $t = \tau$ 时输出达 63.2%， $t = 5\tau$ 时到达稳态。
- **自然频率 $\omega_n = \sqrt{k/m}$** ，**阻尼比 $\xi = \lambda/(2\sqrt{km})$** ，调节时间 $T_s \approx 5/(\xi\omega_n)$ 。
- **工程最优阻尼比 $\xi \approx 0.7$** ：兼顾响应速度与无超调。
- **一阶低通特性**：高频信号被严重衰减，是天然低通滤波器。
- **二阶谐振危机**：当输入频率接近 ω_n 且 ξ 小时发生谐振放大，导致严重测量失真。
- **开环动态补偿**：零极点对消法成本低，但参数漂移时补偿失效（鲁棒性差）。
- **闭环负反馈**：可大幅提升带宽、锁定最优阻尼比、增强鲁棒性。

Tutorial 4: Loading Effects, Noise, Signal Processing & FFT

Q4-1: 负载效应与戴维南等效 (Loading Effect & Thévenin Equivalent)

Potentiometer loading

A potentiometer sensor has $R_P = 1\text{ k}\Omega$, supply voltage $V_S = 10\text{ V}$. A load $R_L = 10\text{ k}\Omega$ is connected. Find the maximum non-linear error.

最大非线性误差公式:

$$\hat{N} \approx \frac{15R_P}{R_L} \% = \frac{15 \times 1000}{10000} \% = 1.5\%$$

为减小非线性误差, 设计时应使 $R_L \gg R_P$ (负载阻抗远大于传感器内阻)。

Q4-2: 噪声与信噪比 (Noise & SNR)

White noise SNR calculation

A sensor signal has RMS voltage $V_S = 1\text{ V}$. The system noise PSD is $\phi = 10^{-8}\text{ W/Hz}$ and bandwidth $f_C = 1000\text{ Hz}$. Calculate the noise RMS voltage and SNR (dB).

$$W_N = \phi \cdot f_C = 10^{-8} \times 1000 = 10^{-5}\text{ W}$$

$$V_{N,\text{rms}} = \sqrt{W_N} = \sqrt{10^{-5}} \approx 3.16\text{ mV}$$

$$\text{SNR} = 20 \log_{10} \frac{V_S}{V_{N,\text{rms}}} = 20 \log_{10} \frac{1}{3.16 \times 10^{-3}} \approx 20 \times 2.50 = 50\text{ dB}$$

Q4-3: 差分放大器共模抑制 (Differential Amplifier CMRR)

CMRR calculation

A differential amplifier has $\text{CMRR} = 10^4$ (80 dB). The common-mode interference is $V_{CM} = 2\text{ V}$. Find the residual series-mode error voltage.

$$V_{CM_error} = \frac{V_{CM}}{\text{CMRR}} = \frac{2}{10^4} = 0.2\text{ mV}$$

Q4-4: 信号平均降噪 (Signal Averaging)

Averaging noise reduction

A sensor signal has noise standard deviation $\sigma = 0.1\text{ V}$. After averaging $p = 100$ repeated measurements, find the reduced noise standard deviation.

$$\sigma_{AV} = \frac{\sigma}{\sqrt{p}} = \frac{0.1}{\sqrt{100}} = \frac{0.1}{10} = 0.01\text{ V}$$

信噪比改善 10 倍。

Q4-5: 奈奎斯特采样定理 (Nyquist Sampling Theorem)

Aliasing prevention

A signal contains components up to $f_{\max} = 5000$ Hz. What is the minimum sampling frequency to avoid aliasing?

$$f_s > 2f_{\max} = 2 \times 5000 = 10000 \text{ Hz} = 10 \text{ kHz}$$

实际工程中通常取 $f_s \geq 2.5f_{\max}$ 以留有裕量。

Tutorial 4 知识点总结

- 负载效应: $V_L = E_{Th} \cdot Z_L / (Z_{Th} + Z_L)$; 设计要求 $Z_L \gg Z_{Th}$ 。
- 电位计非线性误差: $\hat{N} \approx 15R_P/R_L\%$; 设计应使 R_P 小、 R_L 大。
- 白噪声: $W_N = \phi \cdot f_C$, 减小带宽可有效降噪。
- SNR: $\text{SNR}_{dB} = 10 \log_{10}(W_S/W_N) = 20 \log_{10}(V_S/V_N)$ 。
- 共模干扰抑制: $V_{CM_error} = V_{CM}/\text{CMRR}$, CMRR 越高越好。
- 信号平均: $\sigma_{AV} = \sigma/\sqrt{p}$, 适用于周期性重复信号。
- 奈奎斯特定理: $f_s > 2f_{\max}$, 否则发生频谱混叠 (aliasing)。

Tutorial 5: Reliability, TLOC, and Sensing Components

Q5-1: 串并联可靠性计算 (Series/Parallel Reliability)

Mixed redundancy system

Three thermocouples ($\lambda_1 = 1.1 \text{ yr}^{-1}$) are in parallel (median selector), then in series with one converter and one recorder ($\lambda = 0.1 \text{ yr}^{-1}$ each). Find $R_{OVERALL}$ and $F_{OVERALL}$ at $t = 0.5 \text{ yr}$.

步骤 1: 单个热电偶在 $t = 0.5 \text{ yr}$ 的失效概率:

$$F_{\text{single}} = 1 - e^{-1.1 \times 0.5} = 1 - e^{-0.55} \approx 1 - 0.5769 = 0.4231$$

步骤 2: 三并联热电偶组的失效概率 (全部失效才失效):

$$F_1 = (F_{\text{single}})^3 = (0.4231)^3 \approx 0.0757 \implies R_1 = 1 - 0.0757 = 0.9243$$

步骤 3: 变换器与记录仪的可靠度:

$$R_2 = R_3 = e^{-0.1 \times 0.5} = e^{-0.05} \approx 0.9512$$

步骤 4: 整体串联可靠度:

$$R_{OVERALL} = R_1 \times R_2 \times R_3 = 0.9243 \times 0.9512 \times 0.9512 \approx 0.836$$

$$F_{OVERALL} = 1 - 0.836 = 0.164$$

Q5-2: TLOC 全寿命成本比较 (TLOC Analysis)

System TLOC comparison

System 1: $C_1 = \text{£}1000$, $\lambda = 2 \text{ yr}^{-1}$, $C_R = \text{£}20$, $C_L + C_P = \text{£}110/\text{h}$, $T_R = 8 \text{ h}$. System 2: $C_1 = \text{£}2000$, $\lambda = 1 \text{ yr}^{-1}$, $C_R = \text{£}15$, $C_L + C_P = \text{£}110/\text{h}$, $T_R = 12 \text{ h}$. Over $T = 10 \text{ yr}$, which system has lower TLOC?

System 1:

$$\text{Cost/failure} = 20 + 110 \times 8 = \text{£}900$$

$$\text{Total failures} = 2 \times 10 = 20, \quad \text{TLOC}_1 = 1000 + 900 \times 20 = \text{£}19,000$$

System 2:

$$\text{Cost/failure} = 15 + 110 \times 12 = \text{£}1,335$$

$$\text{Total failures} = 1 \times 10 = 10, \quad \text{TLOC}_2 = 2000 + 1335 \times 10 = \text{£}15,350$$

结论: 系统 2 总寿命成本更低 (低 £3,650), 尽管初始购置成本更高。

Q5-3: 电容传感器 (Capacitive Sensor)

Variable dielectric capacitive sensor

A capacitive sensor has plate width $w = 0.02$ m, gap $d = 0.001$ m, plate length $l = 0.05$ m. Dielectric $\varepsilon_1 = 2\varepsilon_0$ slides in by $x = 0.02$ m (air $\varepsilon_2 = \varepsilon_0$). Find the total capacitance change ΔC .

总电容:

$$C_{\text{total}} = \frac{\varepsilon_0 w}{d} [\varepsilon_1 x + \varepsilon_2 (l - x)] = \frac{\varepsilon_0 \times 0.02}{0.001} [2\varepsilon_0 \times 0.02 + \varepsilon_0 \times 0.03]$$

初始 ($x = 0$): $C_0 = \frac{\varepsilon_0 \times 0.02}{0.001} \times \varepsilon_0 \times 0.05 = 20\varepsilon_0^2 \times 0.05$

变化量 (此类变介电常数型传感器, C 与 x 呈严格线性关系):

$$\Delta C = \frac{\varepsilon_0 w}{d} (\varepsilon_1 - \varepsilon_2) x = \frac{\varepsilon_0 \times 0.02}{0.001} \times \varepsilon_0 \times 0.02 = 20\varepsilon_0^2 \times 0.02$$

线性特性是变介电常数型传感器的核心优势。

Q5-4: 可变磁阻传感器 (Variable Reluctance Sensor)

Variable reluctance displacement sensor

An inductive sensor has iron core permeability $\mu_r = 1000$, air gap initial length $\delta_0 = 0.001$ m, cross-section $A = 10^{-4} \text{ m}^2$, coil turns $N = 500$. Find initial air gap reluctance R_{gap} and initial inductance L_0 . (Assume iron core reluctance negligible.)

气隙磁阻 (两侧气隙, 每侧长度 δ_0):

$$R_{\text{gap}} = \frac{2\delta_0}{\mu_0 A} = \frac{2 \times 0.001}{4\pi \times 10^{-7} \times 10^{-4}} = \frac{0.002}{4\pi \times 10^{-11}} \approx 1.592 \times 10^7 \text{ H}^{-1}$$

初始电感:

$$L_0 = \frac{N^2}{R_{\text{gap}}} = \frac{500^2}{1.592 \times 10^7} = \frac{250000}{1.592 \times 10^7} \approx 15.7 \text{ mH}$$

Q5-5: 主动感应元件 — 磁电转速传感器 (Active Inductive Tachometer)

Gear tooth sensor

A gear tachometer has $m = 60$ teeth. The gear rotates at $n = 1200$ RPM. Find the output signal frequency f .

$$f = \frac{n}{60} \times m = \frac{1200}{60} \times 60 = 20 \times 60 = 1200 \text{ Hz}$$

Tutorial 5 知识点总结

- **串联可靠度**：相乘；**并联不可靠度**：相乘（失效概率相乘）。
- **恒定失效率**： $R(t) = e^{-\lambda t}$ ， $\lambda = 1/\text{MTBF}$ 。
- **TLOC**：应综合考虑初始成本、维修频次与停机损失，初始贵但耐用的系统往往全寿命成本更低。
- **电容传感器**：三类变型（变 d 、变 A 、变 ε_r ）；变介电型输出与位移严格线性。
- **磁阻传感器**：气隙磁阻远大于铁芯磁阻；电感与气隙成非线性反比，差动结构可改善线性度并提升灵敏度。
- **磁电转速传感器**：输出频率 $f = (n/60) \times m$ ，幅值与转速成正比。

Tutorial 6: DC Bridge, Linearization & Op-Amp Integration

背景知识

惠斯通电桥输出电压:

$$E_{Th} = V_S \left(\frac{R_1}{R_1 + R_4} - \frac{R_2}{R_2 + R_3} \right)$$

平衡条件: $R_4/R_1 = R_3/R_2$ 。RTD 温度特性: $R_T = R_0(1 + \alpha T)$ 。

Q6-1: 电桥输出计算 (Bridge Output Calculation)

(a) Pt100 RTD bridge

A bridge has $R_2 = R_3 = R_4 = 100 \Omega$, $V_S = 5 \text{ V}$. The RTD $R_1 = R_0(1 + \alpha T)$ where $R_0 = 100 \Omega$, $\alpha = 0.00385 \text{ }^\circ\text{C}^{-1}$. Find E_{Th} at $T = 100 \text{ }^\circ\text{C}$.

$$R_1(100^\circ\text{C}) = 100 \times (1 + 0.00385 \times 100) = 100 \times 1.385 = 138.5 \Omega$$
$$E_{Th} = 5 \left(\frac{138.5}{138.5 + 100} - \frac{100}{100 + 100} \right) = 5 \left(\frac{138.5}{238.5} - 0.5 \right) = 5 \times (0.5807 - 0.5) = 5 \times 0.0807 \approx 0.40 \text{ V}$$

(b) Bridge balance supply voltage

Design the supply voltage to output 100 mV at $T = 100 \text{ }^\circ\text{C}$.

$$\frac{E_{Th}}{V_S} = \left(\frac{138.5}{238.5} - 0.5 \right) = 0.0807 \implies V_S = \frac{0.1}{0.0807} \approx 1.24 \text{ V}$$

Q6-2: 大桥臂比线性化 (Large Bridge Ratio Linearization)

RTD bridge linearization

For the Pt100 bridge above, use bridge ratio $r = R_3/R_2 \gg 1$ approximation. With $r = 10$, find the approximate output at $T = 100 \text{ }^\circ\text{C}$ for $V_S = 5 \text{ V}$.

近似公式:

$$V_{OUT} \approx \frac{V_S}{r} \alpha T = \frac{5}{10} \times 0.00385 \times 100 = 0.5 \times 0.385 = 0.1925 \text{ V}$$

以牺牲灵敏度 (分母增大) 换取良好线性度。

Q6-3: 仪表放大器整合 (Instrumentation Amplifier Integration)

Bridge + differential amplifier

A four-arm active Wheatstone bridge ($R_0 = 120 \Omega$, gauge factor $G = 2.0$) is connected to a differential amplifier ($R_F = 10 \text{ k}\Omega$). Supply $V_S = 15 \text{ V}$. Applied strain $\varepsilon = 500 \mu\epsilon$. Find the final output V_{OUT} .

$$V_{OUT} = 2 \frac{R_F}{R_0} \cdot V_S \cdot G \cdot \varepsilon = 2 \times \frac{10000}{120} \times 15 \times 2.0 \times 500 \times 10^{-6}$$

$$= 2 \times 83.33 \times 15 \times 2.0 \times 5 \times 10^{-4} = 2 \times 83.33 \times 0.015 = 2 \times 1.25 = 2.50 \text{ V}$$

Tutorial 6 知识点总结

- 电桥核心公式: $E_{Th} = V_S(R_1/(R_1 + R_4) - R_2/(R_2 + R_3))$ 。
- 平衡条件: $R_4/R_1 = R_3/R_2$, 此时 $E_{Th} = 0$ 。
- RTD 线性化: 大桥臂比法 $r \gg 1$, 近似公式 $V_{OUT} \approx V_S \alpha T / r$, 以灵敏度换线性度。
- 热敏电阻线性化: 三点设计法, 利用电桥的上曲率抵消热敏电阻的下曲率。
- 运放”虚断”+”虚短”: 理想假设保证电桥公式成立, 且放大倍数仅由反馈电阻决定。
- 四臂全桥 + 差分放大: $V_{OUT} = 2(R_F/R_0)V_S G \varepsilon$, 工程中最高灵敏度方案。

Tutorial 9: Measurement Errors, Statistics & Least Squares

Q9-1: 误差类型与统计量计算 (Error Types & Statistics)

Data statistics

Ten measurements of a resistor (true value $100\ \Omega$): $\{100.2, 99.8, 100.4, 99.6, 100.1, 100.3, 99.9, 100.0, 100.2, 100.5\}\ \Omega$. Find the mean $\bar{x}$, standard deviation S , and systematic error $\bar{E}$.

$$\bar{x} = \frac{1}{10} \sum x_i = \frac{1001.0}{10} = 100.1\ \Omega$$

残差 $v_i = x_i - \bar{x}$: $\{0.1, -0.3, 0.3, -0.5, 0.0, 0.2, -0.2, -0.1, 0.1, 0.4\}$

$$S = \sqrt{\frac{\sum v_i^2}{n-1}} = \sqrt{\frac{0.01 + 0.09 + 0.09 + 0.25 + 0 + 0.04 + 0.04 + 0.01 + 0.01 + 0.16}{9}} = \sqrt{\frac{0.70}{9}} \approx 0.279\ \Omega$$

系统误差: $\bar{E} = \bar{x} - x_{\text{true}} = 100.1 - 100 = +0.1\ \Omega$ (正偏, 需校正)。

Q9-2: 加权平均 (Weighted Average)

Two instruments

Instrument A measures $V_A = 5.10\ \text{V}$ with $\sigma_A = 0.05\ \text{V}$. Instrument B measures $V_B = 5.06\ \text{V}$ with $\sigma_B = 0.02\ \text{V}$. Find the weighted average $\bar{V}$.

权重与方差成反比:

$$p_A = \frac{1}{\sigma_A^2} = \frac{1}{0.0025} = 400, \quad p_B = \frac{1}{\sigma_B^2} = \frac{1}{0.0004} = 2500$$

$$\bar{V} = \frac{p_A V_A + p_B V_B}{p_A + p_B} = \frac{400 \times 5.10 + 2500 \times 5.06}{400 + 2500} = \frac{2040 + 12650}{2900} = \frac{14690}{2900} \approx 5.066\ \text{V}$$

Q9-3: 最小二乘线性回归 (Least Squares Linear Regression)

Sensor calibration

Calibration data for a pressure sensor ($n = 5$ points): x : $\{0, 25, 50, 75, 100\}\ \text{kPa}$; y : $\{0.50, 1.48, 2.53, 3.47, 4.50\}\ \text{V}$. Find the best-fit line $y = a + bx$.

计算各统计量:

$$\bar{x} = 50, \quad \bar{y} = \frac{0.50 + 1.48 + 2.53 + 3.47 + 4.50}{5} = \frac{12.48}{5} = 2.496$$

$$S_{xx} = \sum (x_i - \bar{x})^2 = 50^2 + 25^2 + 0 + 25^2 + 50^2 = 2500 + 625 + 0 + 625 + 2500 = 6250$$

$$\begin{aligned} S_{xy} &= \sum (x_i - \bar{x})(y_i - \bar{y}) = (-50)(0.50 - 2.496) + (-25)(1.48 - 2.496) + 0 + 25 \times 0.974 + 50 \times 2.004 \\ &= (-50)(-1.996) + (-25)(-1.016) + 0 + 24.35 + 100.2 = 99.8 + 25.4 + 24.35 + 100.2 = 249.75 \end{aligned}$$

斜率（灵敏度）：

$$b = \frac{S_{xy}}{S_{xx}} = \frac{249.75}{6250} \approx 0.0400 \text{ V/kPa}$$

截距（零偏）：

$$a = \bar{y} - b\bar{x} = 2.496 - 0.040 \times 50 = 2.496 - 2.0 = 0.496 \text{ V}$$

最佳拟合线： $y = 0.496 + 0.040x$ 。

Q9-4: 传感器静态特性综合评估 (Static Characteristics)

Linearity and hysteresis errors

A sensor with full-scale output $y_{FS} = 10 \text{ V}$ has maximum non-linear deviation $\Delta L_{\max} = 0.15 \text{ V}$ and maximum hysteresis difference $\Delta H_{\max} = 0.08 \text{ V}$. Calculate the errors.

$$\gamma_L = \pm \frac{\Delta L_{\max}}{y_{F.S.}} \times 100\% = \pm \frac{0.15}{10} \times 100\% = \pm 1.5\%$$
$$\gamma_H = \pm \frac{\Delta H_{\max}}{y_{F.S.}} \times 100\% = \pm \frac{0.08}{10} \times 100\% = \pm 0.8\%$$

Tutorial 9 知识点总结

- 三类误差：系统误差（有规律，需校正）、随机误差（统计规律，取均值）、粗大误差（ 3σ 准则剔除）。
- 标准差 Bessel 公式： $S = \sqrt{\sum v_i^2 / (n - 1)}$ ，评估测量离散程度。
- 加权平均：精度高的仪器权重大（ $p_i = 1/\sigma_i^2$ ）。
- 最小二乘法： $b = S_{xy}/S_{xx}$ ， $a = \bar{y} - b\bar{x}$ ，用于传感器静态标定。
- 线性度误差 $\gamma_L = \pm \Delta L_{\max}/y_{FS} \times 100\%$ ；迟滞误差 $\gamma_H = \pm \Delta H_{\max}/y_{FS} \times 100\%$ 。
- 动态特性：一阶 τ 越小越快；二阶 $\xi \approx 0.707$ 为工程最优。

Tutorial 10: Inductive & Capacitive Sensors, Strain Gauges

Q10-1: 差动电感传感器 (Differential Inductive Sensor)

Differential inductance calculation

An inductive sensor has $W = 2000$ turns, $S_0 = 4 \text{ cm}^2$, $\delta_0 = 0.2 \text{ cm}$. Find initial inductance L_0 and differential sensitivity K_{diff} .

单位换算: $S_0 = 4 \times 10^{-4} \text{ m}^2$, $\delta_0 = 0.2 \times 10^{-2} = 2 \times 10^{-3} \text{ m}$ 。

$$\begin{aligned} L_0 &= \frac{\mu_0 S_0 W^2}{2\delta_0} = \frac{4\pi \times 10^{-7} \times 4 \times 10^{-4} \times 4 \times 10^6}{2 \times 2 \times 10^{-3}} = \frac{4\pi \times 10^{-7} \times 1.6 \times 10^3}{4 \times 10^{-3}} \\ &= \frac{4\pi \times 1.6 \times 10^{-4}}{4 \times 10^{-3}} = \pi \times 1.6 \times 10^{-4} / 10^{-3} \approx 0.503 \text{ H} \end{aligned}$$

单端灵敏度: $K_{\text{single}} \approx L_0 / \delta_0 = 0.503 / (2 \times 10^{-3}) \approx 251 \text{ H/m}$

差动灵敏度: $K_{\text{diff}} = 2K_{\text{single}} \approx 502 \text{ H/m}$ (比单端提高一倍)

Q10-2: 圆柱形电容液位传感器 (Cylindrical Capacitive Level Gauge)

Liquid level sensor

A cylindrical capacitive sensor has inner radius $r_1 = 0.05 \text{ m}$, outer radius $r_2 = 0.06 \text{ m}$, total height $H = 1 \text{ m}$, liquid dielectric $\varepsilon_r = 2.5$. Find the sensitivity $\Delta C / \Delta h$.

圆柱电容公式: $C = \frac{2\pi\varepsilon_0\varepsilon_r h}{\ln(r_2/r_1)}$

灵敏度 (ΔC 相对于高度 h):

$$\begin{aligned} \frac{\Delta C}{\Delta h} &= \frac{2\pi\varepsilon_0(\varepsilon_r - 1)}{\ln(r_2/r_1)} = \frac{2\pi \times 8.854 \times 10^{-12} \times (2.5 - 1)}{\ln(0.06/0.05)} = \frac{2\pi \times 8.854 \times 10^{-12} \times 1.5}{\ln(1.2)} \\ &= \frac{83.47 \times 10^{-12}}{0.1823} \approx 458 \text{ pF/m} \end{aligned}$$

Q10-3: 应变片全桥回路 (Full-Bridge Strain Gauge)

Cantilever beam full bridge

A cantilever beam has strain gauge $G = 2.1$, $R = 120 \Omega$, supply $U_e = 10 \text{ V}$. Applied force produces $\varepsilon = 800 \mu\varepsilon$. Full bridge configuration. Find output voltage U_o .

全桥输出公式:

$$U_o = U_e \cdot G \cdot \varepsilon = 10 \times 2.1 \times 800 \times 10^{-6} = 10 \times 2.1 \times 8 \times 10^{-4} = 16.8 \times 10^{-3} = 16.8 \text{ mV}$$

Tutorial 10 知识点总结

- 电感传感器: $L_0 = \mu_0 S_0 W^2 / (2\delta_0)$; 差动结构灵敏度翻倍且线性度改善。
- 电容传感器: 变介电型最线性 ($C \propto h$ 或 $C \propto x$); 变距型最非线性。
- 应变片转换链: 力 $\rightarrow$ 应力 $\sigma = F/A \rightarrow$ 应变 $\varepsilon = \sigma/E \rightarrow \Delta R/R = G\varepsilon \rightarrow$ 电压输出。
- 桥路配置比较: 四分之一桥 (1 倍)、半桥差动 (2 倍)、全桥差动 (4 倍), 且差动配置同时消除非线性和温度误差。

Tutorial 11: Strain Gauges — Advanced Applications

Q11-1: 纵向与横向应变片 (Longitudinal & Transverse Gauges)

Poisson effect and transverse gauge

A steel bar ($E = 200 \text{ GPa}$, $\nu = 0.28$) carries axial compressive load $F = 50 \text{ kN}$. Cross-section $A = 500 \text{ mm}^2$. $G = 2.0$, $R = 120 \Omega$. Find (a) longitudinal strain ε_L ; (b) transverse strain ε_T ; (c) $\Delta R/R$ for longitudinal gauge; (d) $\Delta R/R$ for transverse gauge.

(a) 纵向应变:

$$\sigma = \frac{F}{A} = \frac{-50 \times 10^3}{500 \times 10^{-6}} = -10^8 \text{ Pa} = -100 \text{ MPa}$$
$$\varepsilon_L = \frac{\sigma}{E} = \frac{-10^8}{200 \times 10^9} = -5 \times 10^{-4} = -500 \mu\varepsilon$$

(b) 横向应变 (泊松效应):

$$\varepsilon_T = -\nu\varepsilon_L = -0.28 \times (-500) = +140 \mu\varepsilon$$

(横向膨胀, 为正值)

(c) 纵向应变片电阻变化率:

$$\frac{\Delta R}{R} = G\varepsilon_L = 2.0 \times (-500 \times 10^{-6}) = -1.0 \times 10^{-3} = -0.1\%$$

(d) 横向应变片电阻变化率:

$$\frac{\Delta R}{R} = G\varepsilon_T = 2.0 \times 140 \times 10^{-6} = +2.8 \times 10^{-4}$$

Q11-2: 悬臂梁桥路设计 (Cantilever Beam Bridge Design)

Bridge wiring for cantilever beam

A cantilever beam has gauges on top (tension, $+\varepsilon$) and bottom (compression, $-\varepsilon$). Design a half-bridge configuration and write the output formula.

接线原则: ”一拉一压接相邻臂”——上表面拉伸片 $R_1 (+\Delta R)$ 与下表面压缩片 $R_2 (-\Delta R)$ 接在相邻桥臂。

半桥输出公式 (精确):

$$U_o = \frac{R_1 + \Delta R}{(R_1 + \Delta R) + (R_2 - \Delta R)} - \frac{1}{2} = \frac{R + \Delta R}{2R} - \frac{1}{2} = \frac{\Delta R}{2R}$$
$$\therefore U_o = \frac{1}{2} \cdot \frac{\Delta R}{R} \cdot U_i = \frac{1}{2} \cdot G\varepsilon \cdot U_i$$

半桥差动消除了非线性误差, 灵敏度是四分之一桥的 2 倍。

Q11-3: 柱式力测量系统 (Column-type Force Measurement)

Column force sensor with Poisson effect

A steel column ($\nu = 0.3$, $E = 200 \text{ GPa}$, $A = 0.01 \text{ m}^2$) measures compressive force. 4 longitudinal gauges and 4 transverse gauges in full bridge. $G = 2.0$, $R = 120 \Omega$, $U_i = 10 \text{ V}$. Applied force $F = 500 \text{ kN}$. Find output U_o .

轴向压应变:

$$\varepsilon_L = \frac{\sigma}{E} = \frac{-F/A}{E} = \frac{-500 \times 10^3 / (0.01)}{200 \times 10^9} = \frac{-5 \times 10^7}{2 \times 10^{11}} = -2.5 \times 10^{-4}$$

横向应变 (拉): $\varepsilon_T = -\nu\varepsilon_L = 0.3 \times 2.5 \times 10^{-4} = +7.5 \times 10^{-5}$

跨维度差动全桥输出:

$$U_o = 2(1 + \nu) \cdot G \cdot |\varepsilon_L| \cdot \frac{U_i}{4} \times 4$$

精确利用公式:

$$\begin{aligned} U_o &\approx \frac{2(1 + \nu)\Delta R_{\text{long}}}{4R} \cdot U_i \times 4 = (1 + \nu) \cdot G \cdot |\varepsilon_L| \cdot U_i \\ &= (1 + 0.3) \times 2.0 \times 2.5 \times 10^{-4} \times 10 = 1.3 \times 5 \times 10^{-3} = 6.5 \text{ mV} \end{aligned}$$

经仪表放大器 (如增益 $\times 200$) 后: $V_{\text{final}} = 6.5 \text{ mV} \times 200 = 1.3 \text{ V}$ 。

Tutorial 11 知识点总结

- 泊松效应: $\varepsilon_T = -\nu\varepsilon_L$, 符号相反 (拉伸时横向缩短, 压缩时横向膨胀)。
- 柱式传感器突破: 纵向片 (压缩) 与横向片 (拉伸) 交替入桥, 实现”跨维度”差动全桥, 同时消除非线性误差与温度漂移。
- 全桥输出放大倍数: $U_o = (1 + \nu)G|\varepsilon_L|U_i$ 约为四分之一桥的 $2(1 + \nu) \approx 2.6$ 倍。
- 桥路设计黄金法则: 相邻桥臂应变方向相反 (信号叠加); 对角桥臂应变方向相同 (信号也叠加)。
- 信号调理: 桥路输出为毫伏级, 必须通过高 CMRR 仪表放大器进行级联放大才能驱动 ADC。

总总结 (Global Summary) — Sensor-Integrated Systems Tutorials 1–11

本系列教程涵盖传感器集成系统的完整知识体系，核心脉络如下：

一、静态特性体系 (Tutorials 1, 9)

- 理想直线 $O = KI + a$ ，非线性误差、干扰/修正输入构成完整静态模型。
- 最小二乘法用于静态标定； 3σ 准则剔除粗大误差。

二、误差分析与系统架构 (Tutorials 2, 9)

- 串联系统误差级级放大；高增益负反馈将精度“锁定”于反馈元件，大幅提升鲁棒性。
- 双轨道算法：均值误差用均值代入，随机误差用偏导数传播法。

三、动态响应 (Tutorial 3)

- 一阶：时间常数 τ ；二阶：自然频率 ω_n 、阻尼比 ξ （工程最优 ≈ 0.7 ）。
- 开环补偿（零极点对消）成本低但脆弱；闭环反馈带宽大且鲁棒。

四、信号质量与处理 (Tutorial 4)

- 负载效应： $Z_L \gg Z_{Th}$ 。噪声：减小带宽、信号平均、差分放大是三大抑制手段。
- 奈奎斯特定理： $f_s > 2f_{\max}$ ，FFT 将时域“乱麻”变频域“清单”。

五、可靠性与工程经济 (Tutorial 5)

- 串联乘可靠度，并联乘不可靠度。TLOC 综合比较：初始贵但耐用的系统往往全寿命成本更低。

六、传感器物理机制 (Tutorials 5, 10, 11)

- 电容： $C = \epsilon_0 \epsilon_r A/d$ ；变介电型线性最佳。
- 电感：气隙磁阻主导，差动结构线性度翻倍、灵敏度翻倍。
- 应变片：力 $\rightarrow$ 应力 $\rightarrow$ 应变 $\rightarrow \Delta R/R = G\epsilon$ ；泊松效应使柱式传感器可构成“跨维度”差动全桥。

七、信号调理电路 (Tutorials 6, 10, 11)

- 惠斯通电桥将微小电阻变化转为可测电压。大桥臂比法/三点法改善线性度。
- 运放“虚断虚短”是分析桥路 + 放大器的两把钥匙。
- 工程完整链路：传感物理量 $\rightarrow$ 桥路电压 (mV 级) $\rightarrow$ 仪表放大器 ($\rightarrow$ V 级) $\rightarrow$ ADC $\rightarrow$ 数字处理。